

LUND UNIVERSITY

Mathematics, Faculty of Science

Number theory
MATM15
HT 2019
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Written examination
January 9, 2020, 8:00-13:00

1. Show that 2 is a primitive root modulo 19. Compute a corresponding table of indices. Use this to solve the congruence

$$3x^{15} \equiv 2 \pmod{19}.$$

2. Prove that

$$\sum_{d|n} 2^{\omega(d)} = \tau(n^2),$$

where τ is the divisor function and $\omega(n)$ is the number of distinct prime factors of n.

3. Given the continued fraction expansion $\sqrt{24} = [4; \overline{1, 8}]$, find a convergent p_n/q_n that satisfies

$$\left| \sqrt{24} - \frac{p_n}{q_n} \right| < \frac{1}{1000}.$$

4. Let a and b be positive integers and $d = \gcd(a, b)$. Prove that

$$\phi(d)\phi(ab) = d\phi(a)\phi(b),$$

where ϕ is the Euler ϕ-function.

5. Find the smallest prime p such that 90 is a quadratic residue modulo p.

6. Let q be a prime.
   (a) If a is a positive integer such that $q \mid a$, and p is a prime such that

$$p \mid (1 + a + \cdots + a^{q-1}),$$

   prove that $p \equiv 1 \pmod{q}$.
   (b) Prove that there are infinitely many primes congruent to 1 modulo q.

Give careful motivations for your solutions!